Exploratory Data Analysis

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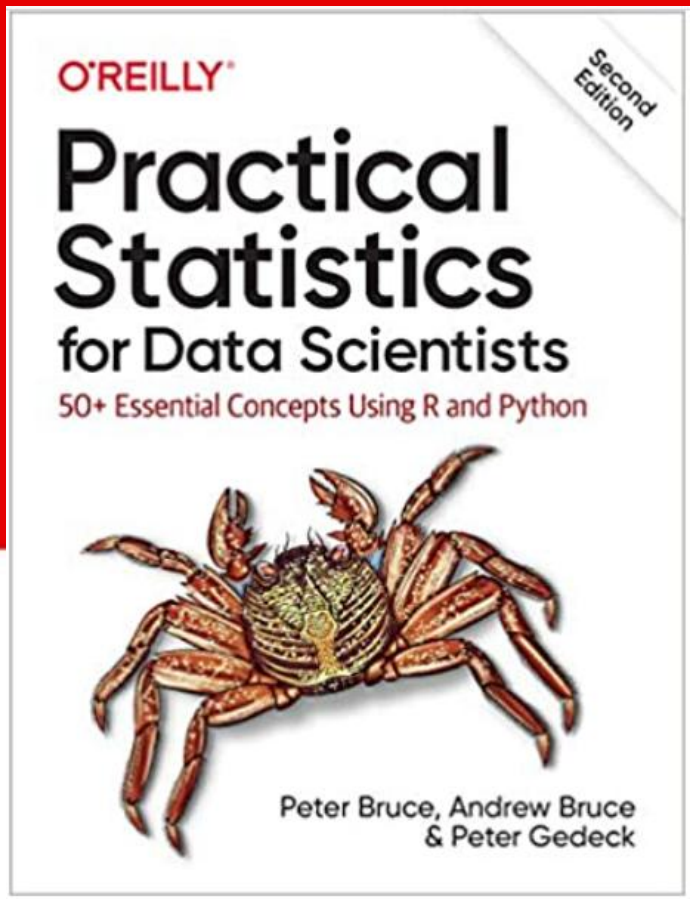
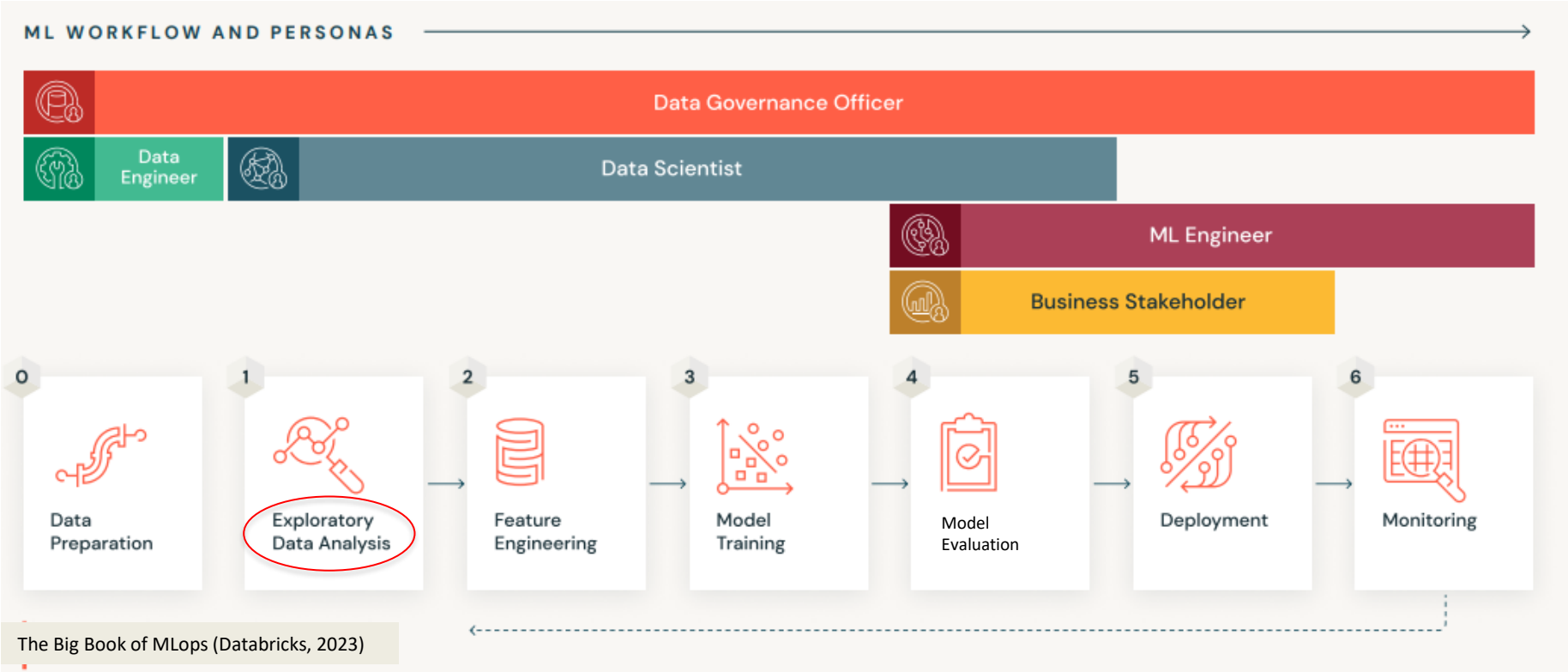
CHAPTER 1

EXPLORATORY DATA ANALYSIS

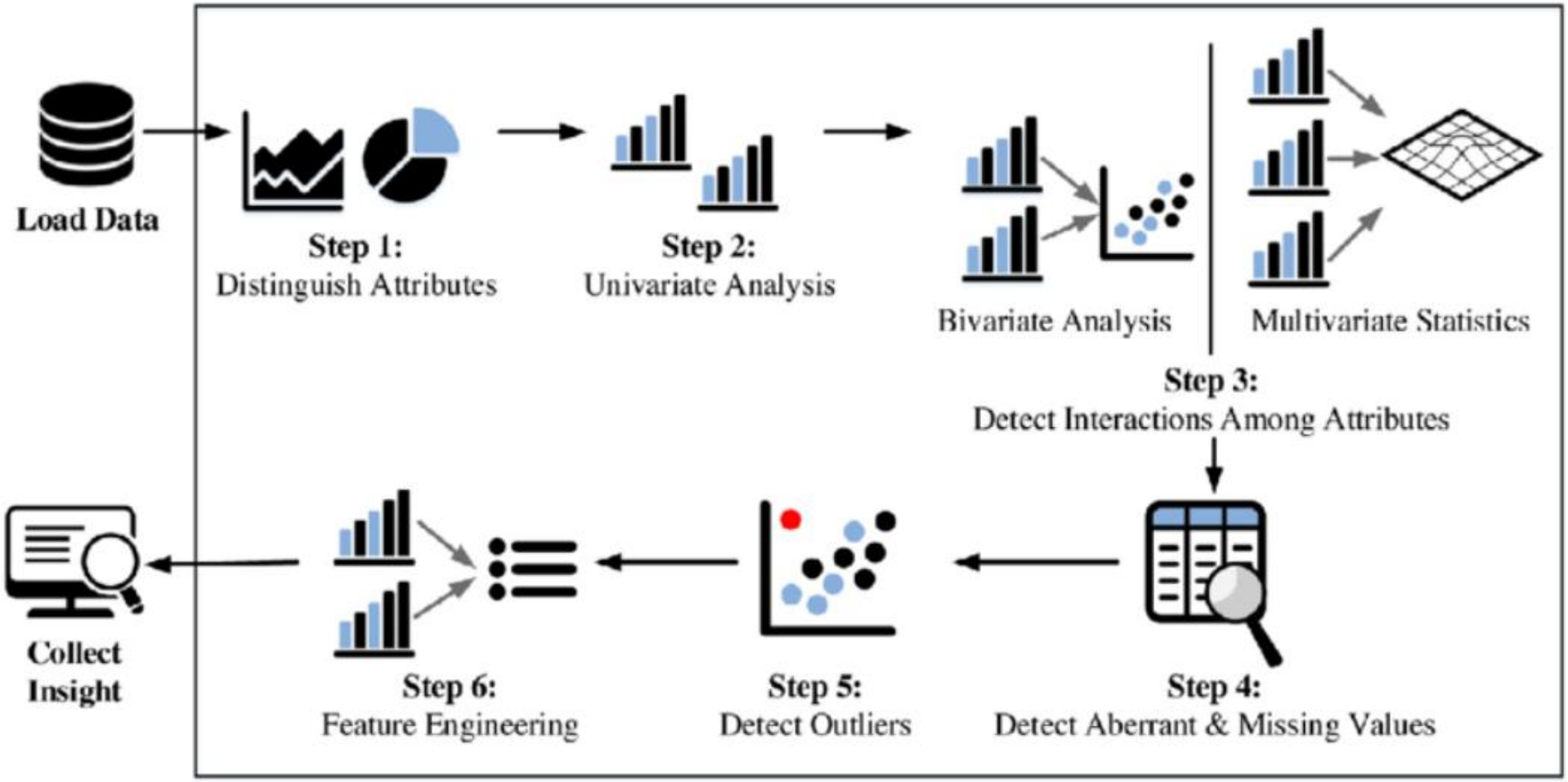


The Machine Learning workflow



The Big Book of MLops (Databricks, 2023)

Exploratory Data Analysis (EDA)



Source: https://doi.org/10.1016/j.visinf.2018.12.004

Origins of EDA

- **John W. Tukey** (1962) proposed data analysis as a broader field beyond classical statistical inference
- He linked statistics with engineering and computer science (he also coined "bit" and "software").
- He published the book "Exploratory Data Analysis" (1977), promoting simple plots and summary statistics.
- EDA expanded with advances in technology, bigger data, and better analysis tools.
- Tukey's work laid the foundation for modern data science (according to David Donoho, 2015).

"The greatest value of a picture is when it forces us to notice what we never expected to see." — John W. Tukey

What is EDA?

- **Exploratory Data Analysis is the process of understanding the dataset's structure, relationships, and patterns before modeling.**

- **Why perform EDA?**
  - Clean and prepare data
  - Visualize relationships
  - Optimize feature selection
  - Avoid incorrect conclusions

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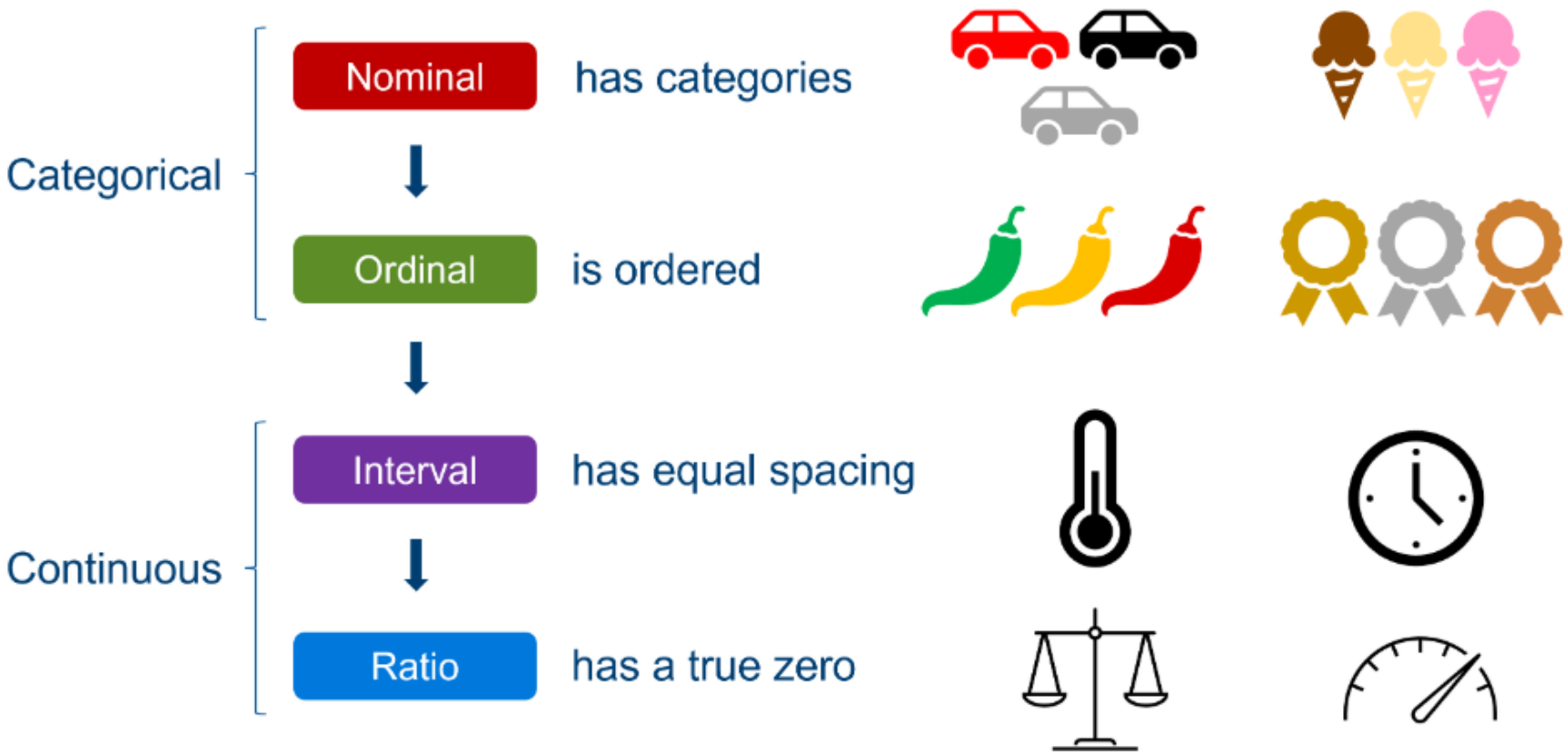
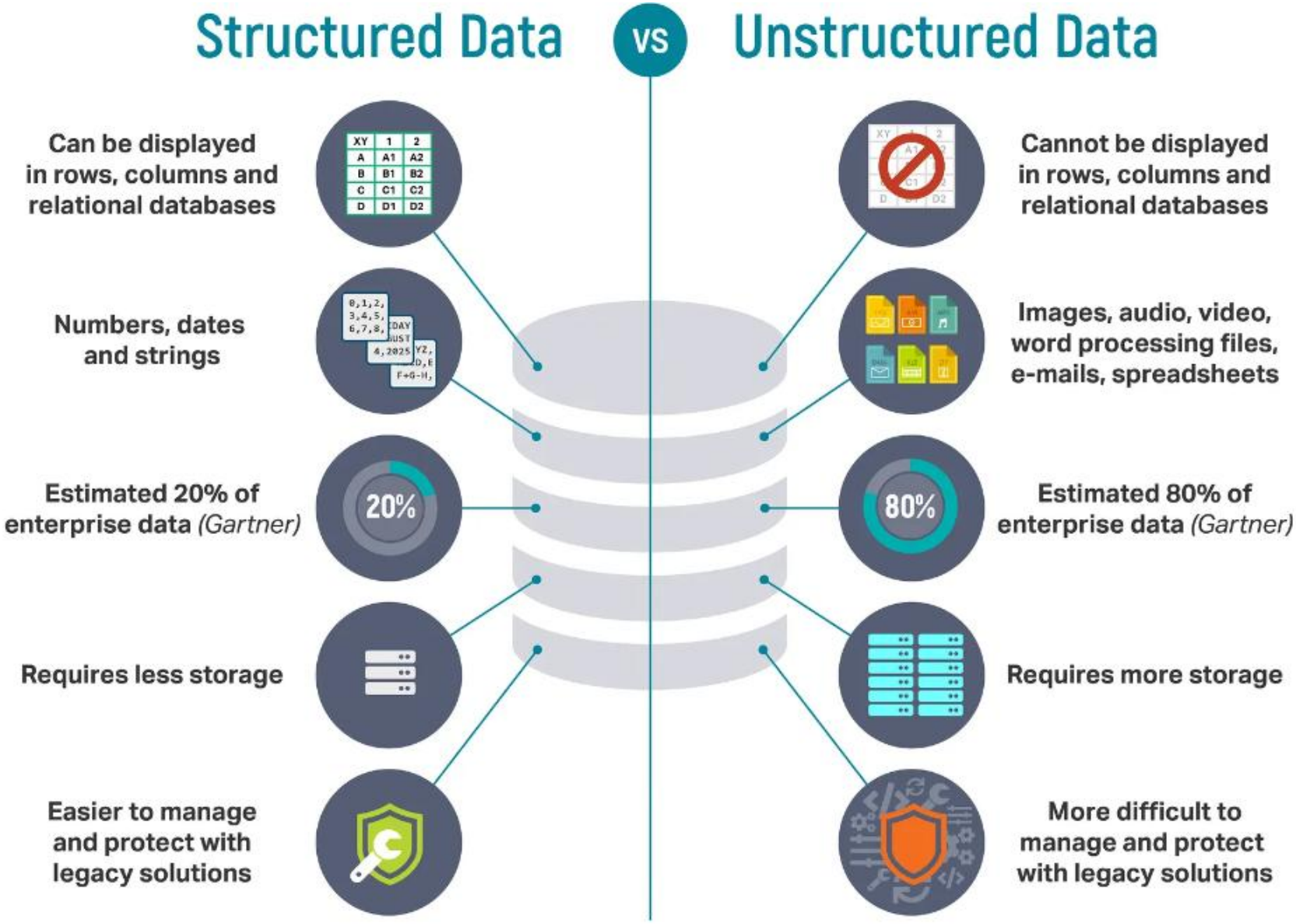
Different types of data



Source: https://cambiotraining.github.io/stats-week/materials/variables-and-data-types.html

Analysis based on variable types

- **Nominal**: frequency counts, mode, bar charts, pie charts, …
- **Ordinal**: median, percentiles, IQR, box plots, …
- **Quantitative**: mean, median, standard deviation, histograms, …



https://www.igneous.io/blog/structured-data-vs-unstructured-data

Rectangular data

= the basic data structure in **data science**

- **Dataframe** = table = dataset
- **Example** = sample = instance = data point = observation = case = pattern = record = table row
- **Feature** = independent variable = input = predictor = attribute = table column
- **Target** = dependent variable = output = outcome = response = feature we want to predict

data(set)

	A	B	C	D	E	F
1	id	date	size	typos	recipients	spam
2	0	12/01/2021	2.5	0	1	False
3	1	13/01/2021	1.3	0	2	False
4	2	14/01/2021	12.1	3	15	True
5	3	15/01/2021	7.8	2	19	True
6	4	16/01/2021	4.6	1	5	False
7	5	17/01/2021	9.8	5	1	True
8	6	18/01/2021	11.6	3	63	True

example

feature target

Rectangular data

= the basic data structure in **data science**

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X = feature matrix
= input variables

y = target vector
= output variable

X y

data(set)

	A	B	C	D	E	F
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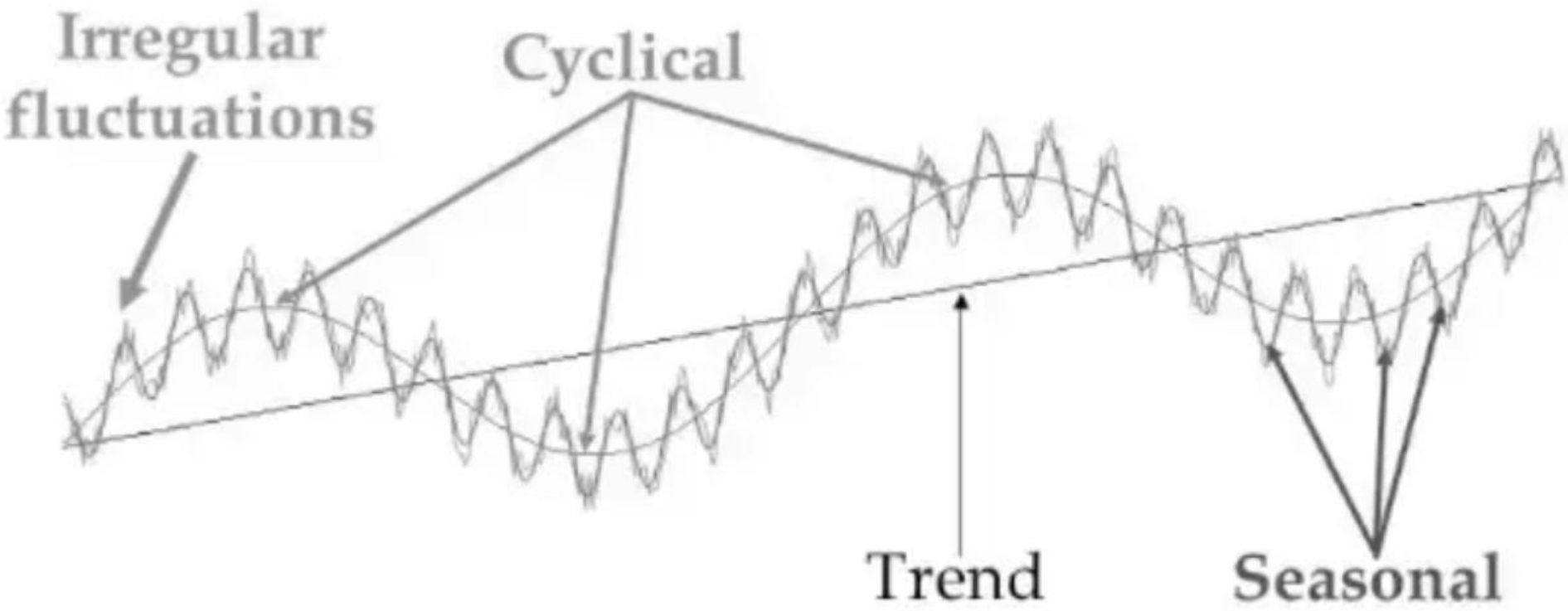
example

feature target

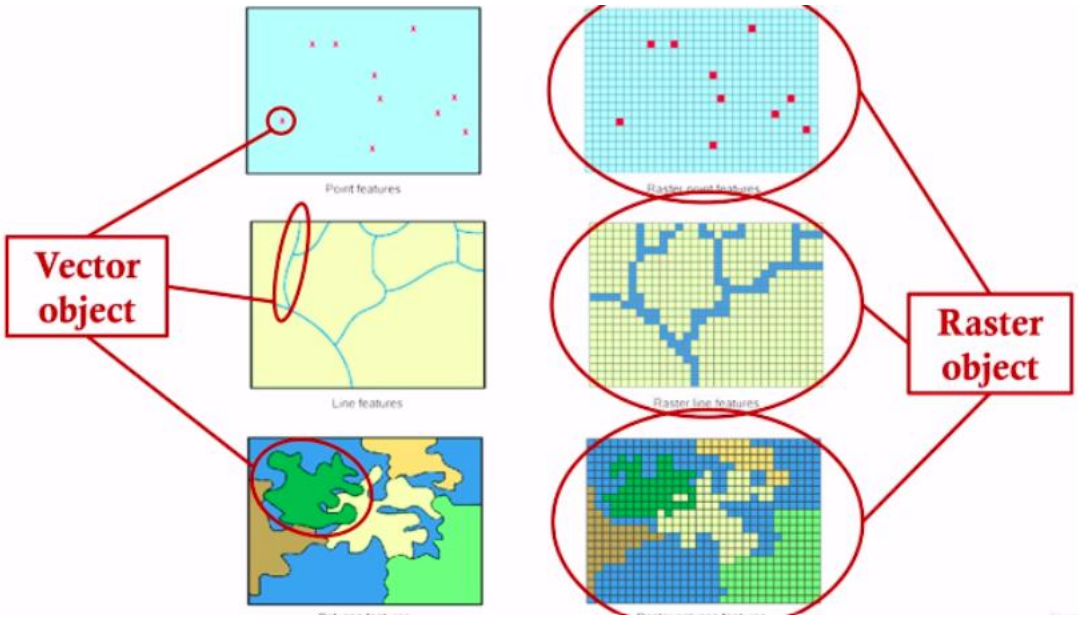
Time series

- Time series data records successive measurements of the same variable.
- Time series has <u>time as index</u>
- E.g., data produced by IoT devices

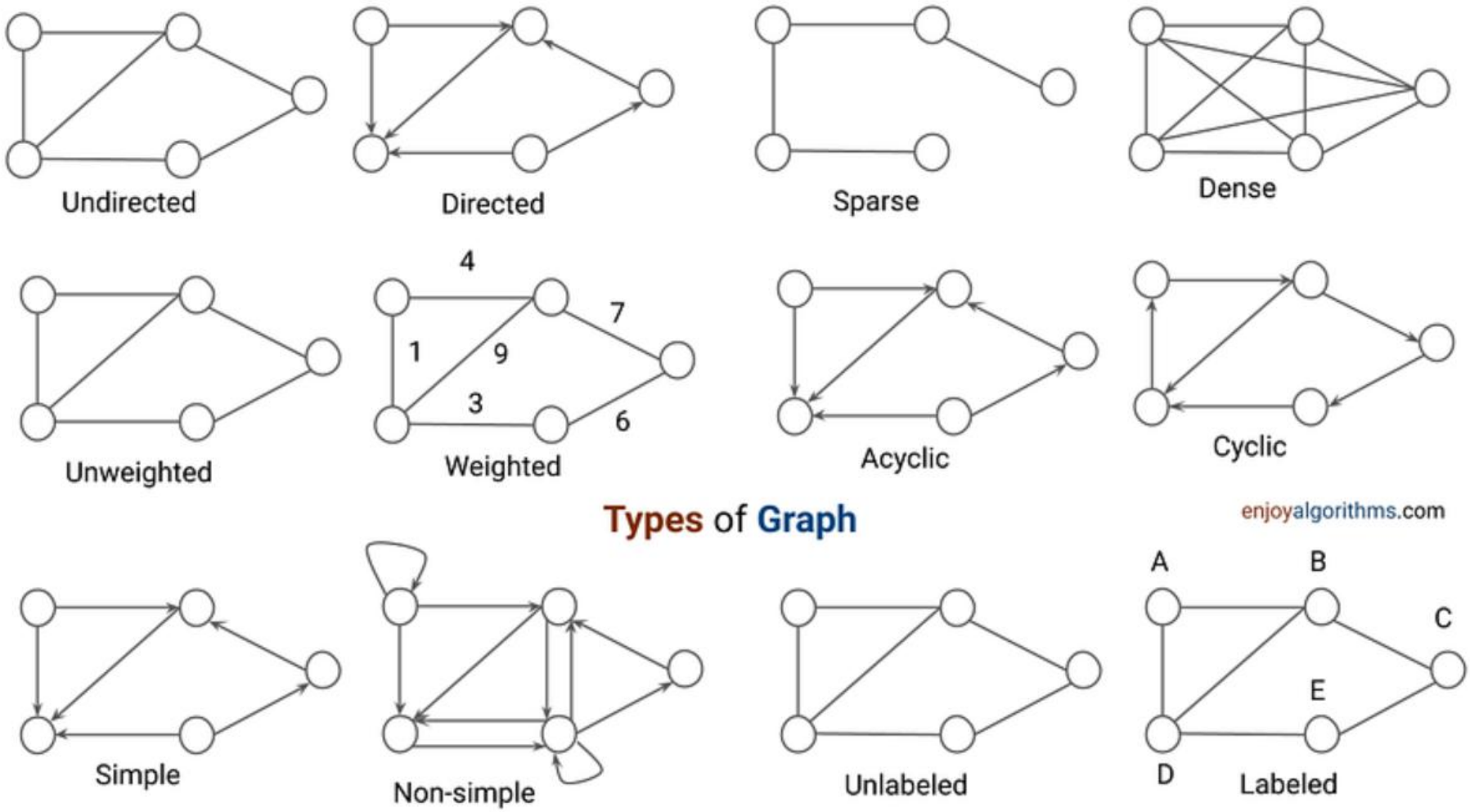


(Geo)spatial data

- refers to information about the **physical location** and shape of objects, often represented in a **coordinate system**.
- can include details about natural features like mountains and rivers, as well as human-made structures like buildings and roads.
- **Geographic Information Systems** (GIS) are tools designed to capture, store, analyze, manage, and visualize spatial data.



Graph data structures



Source: https://www.enjoyalgorithms.com/blog/types-of-graphs-in-data-structures

Estimates of location

- Mean = average
- Weighted mean
- Median = 50th percentile
- Weighted median
- Percentile = quantile
- Trimmed mean = truncated mean
- …

- **Robust** = not sensitive to extreme values (outliers):

  e.g., median, percentiles, and trimmed mean

Estimates of variability

- Standard deviation
- Variance
- Mean absolute deviation = mean absolute error (MAE)
- Mean squared deviation = mean squared error (MSE)
- Median absolute deviation (MAD)
- Differences between percentiles
- Interquartile range (IQR)
- …

- **Robust:** MAE, MSE, MAD, IQR, …

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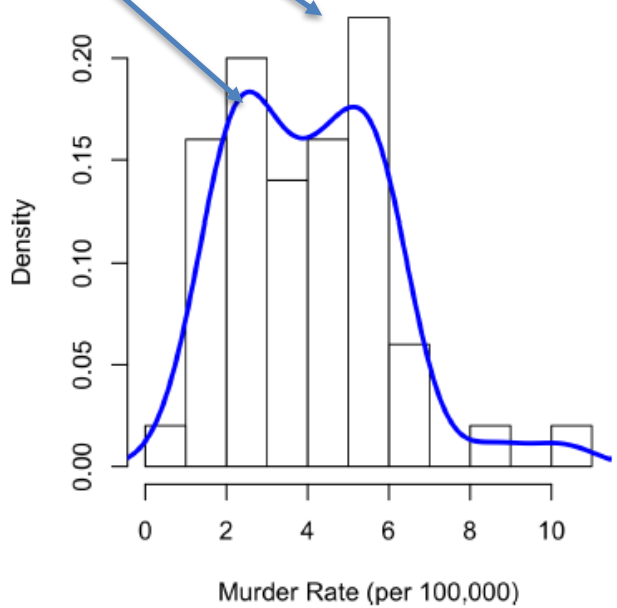
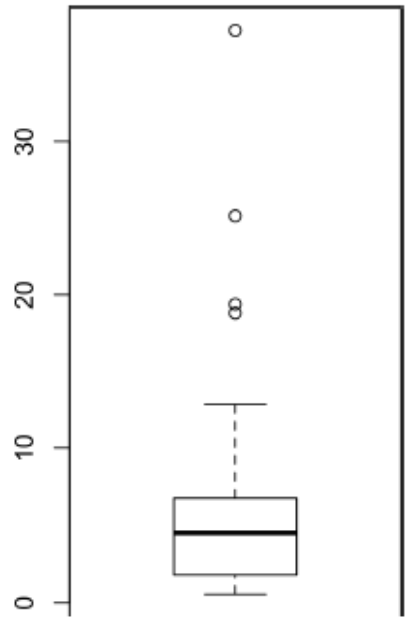
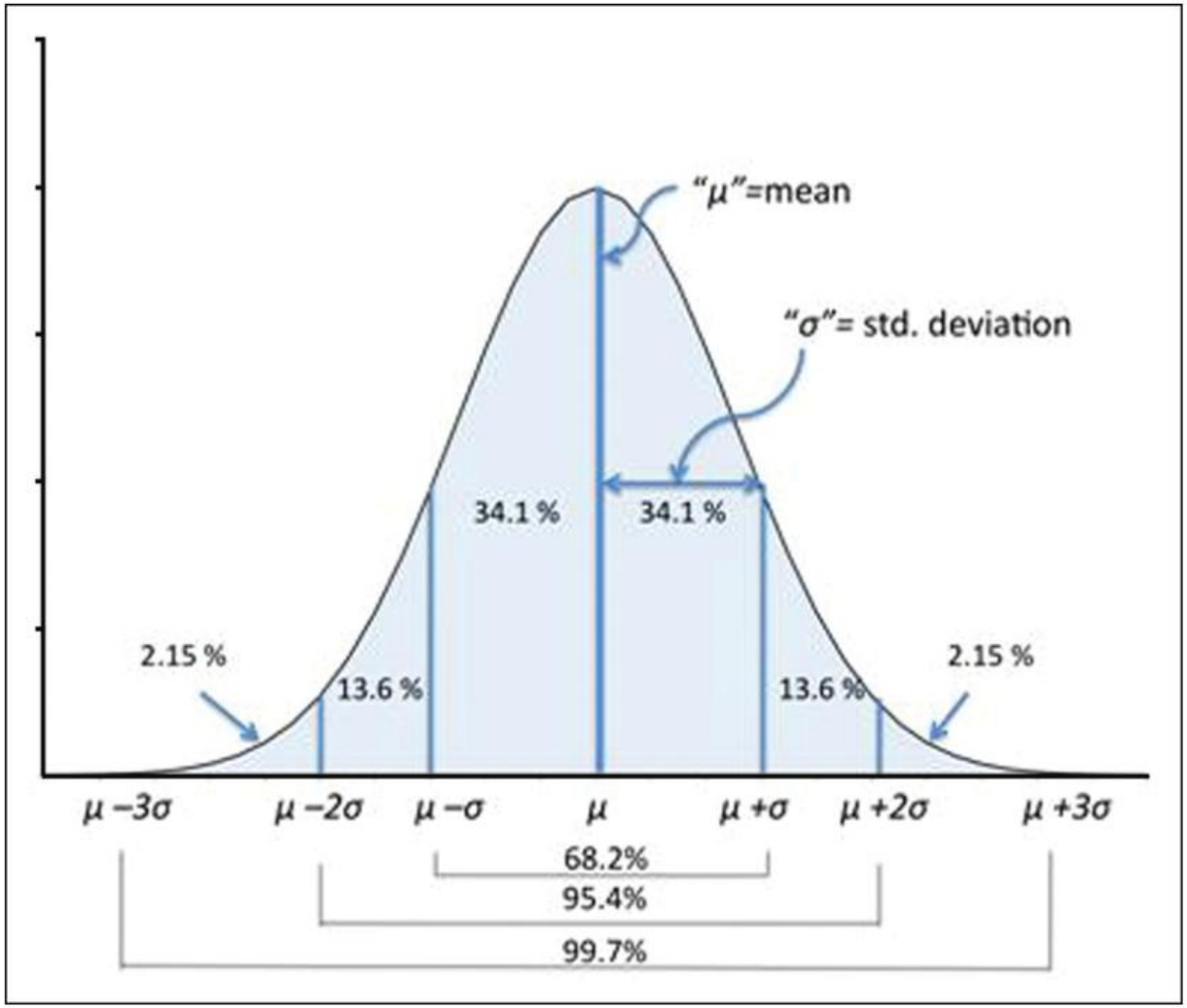
Exploring the data distribution

- Boxplot
- Frequency table
- Histogram
- Density plot



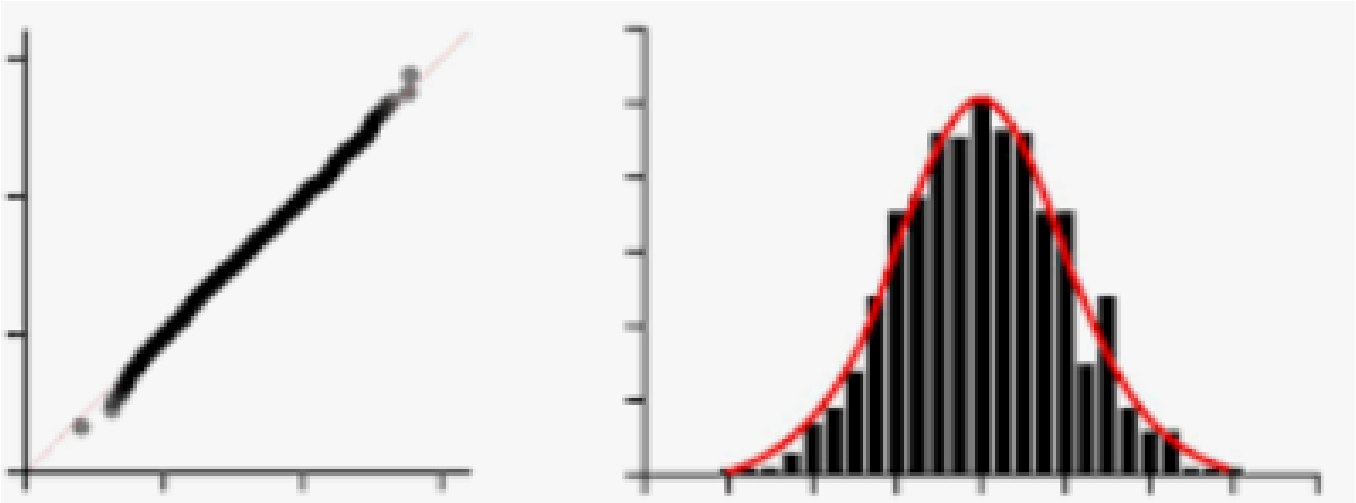


Normal distribution

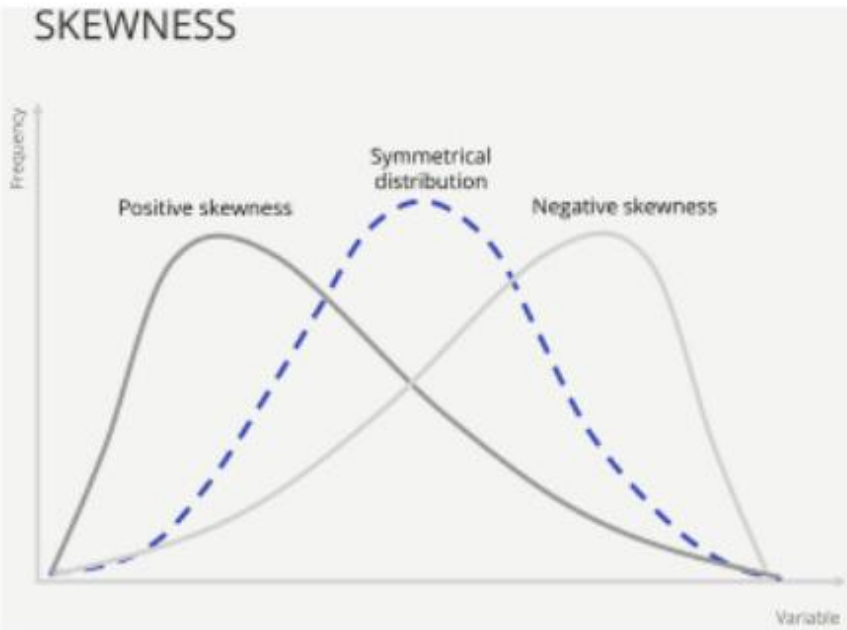


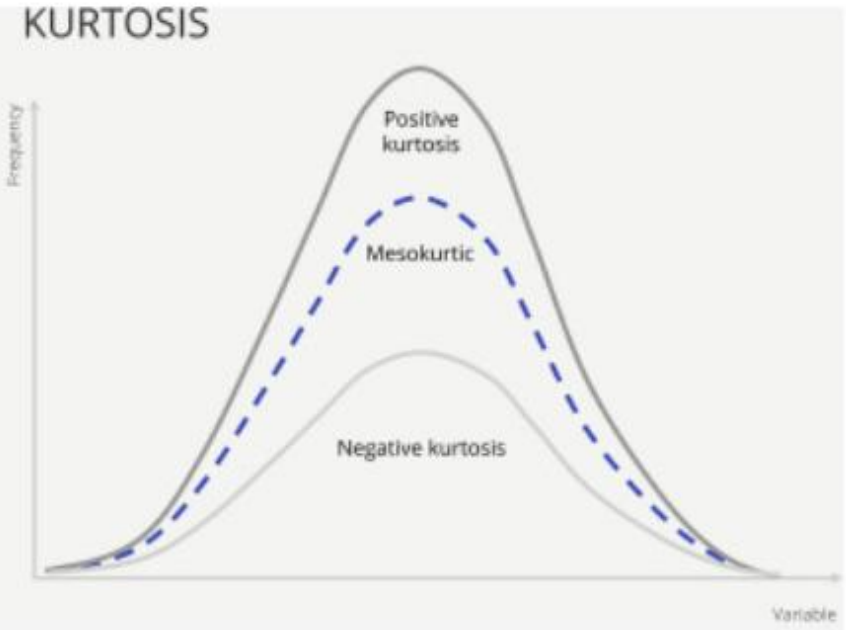
Testing for normality

- Statistical tests (e.g. Shapiro-Wilk)
- Visual methods: Q-Q plot, histogram



Skewness and kurtosis

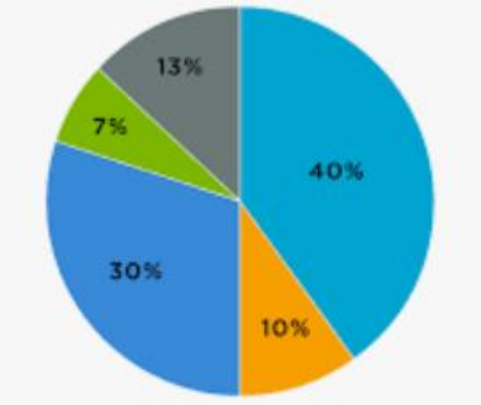


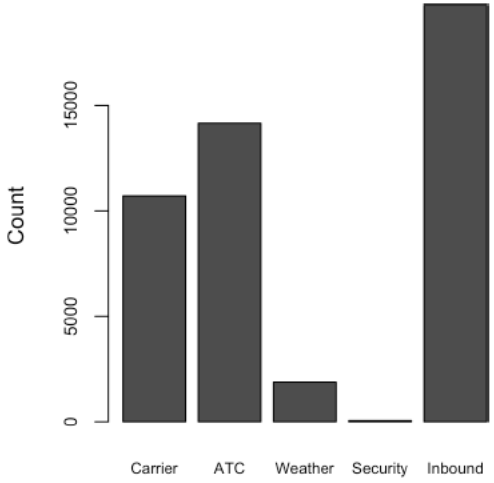


Exploring binary and categorical data

- Counts
- Mode = most commonly occurring category

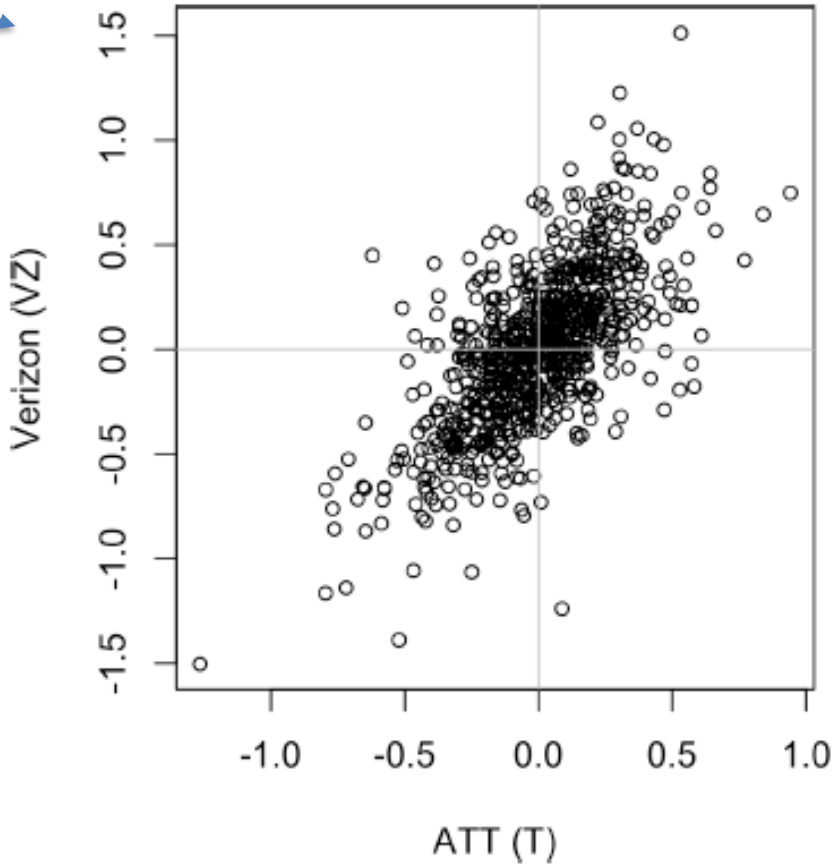
- Bar charts
- Pie charts





Correlation

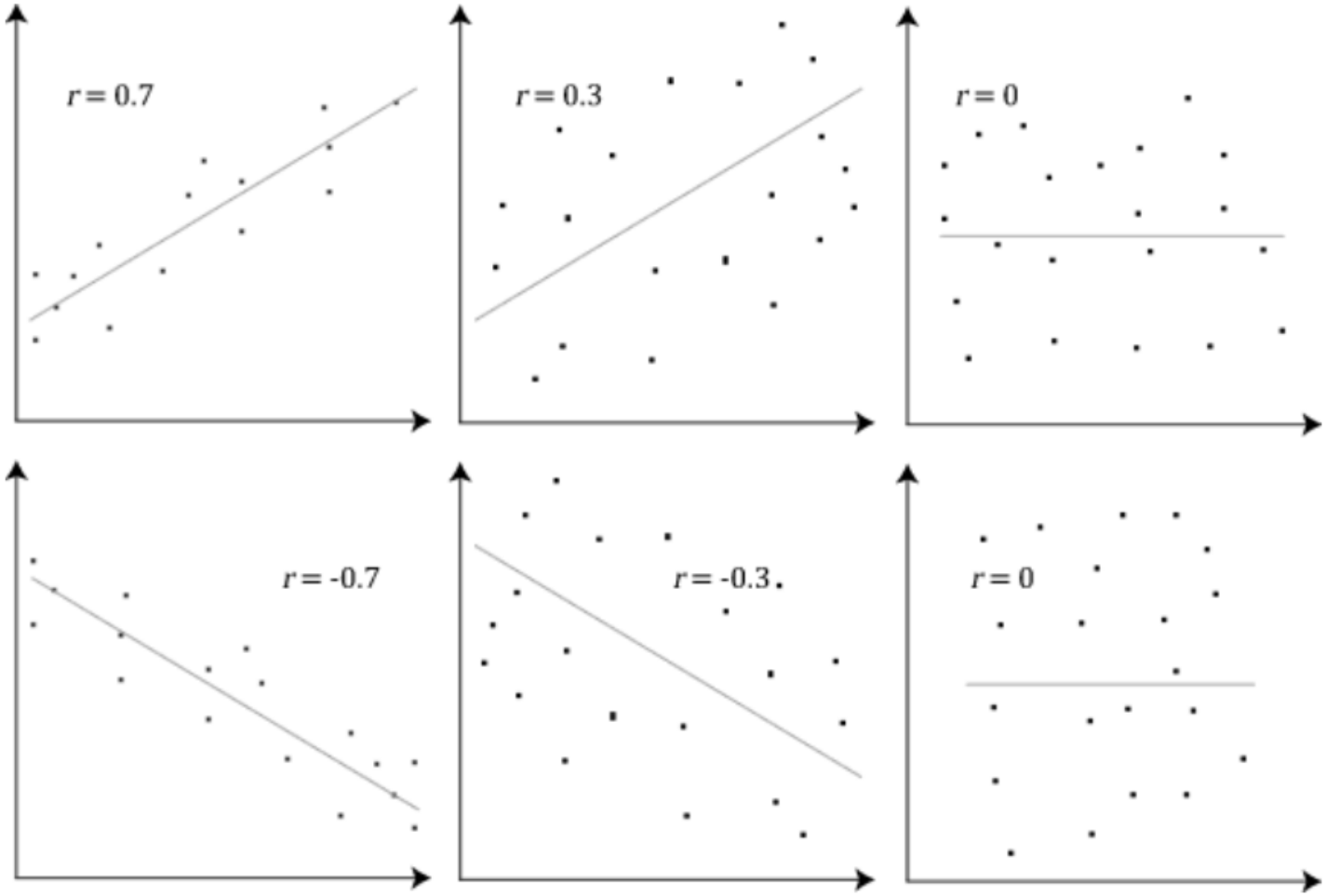
- Scatter plots



- Correlation coefficient
- Correlation matrix

- Quantitative variables!

Pearson's Correlation Coefficient



Source: https://statistics.laerd.com/statistical-guides/pearson-correlation-coefficient-statistical-guide.php

Correlation matrix

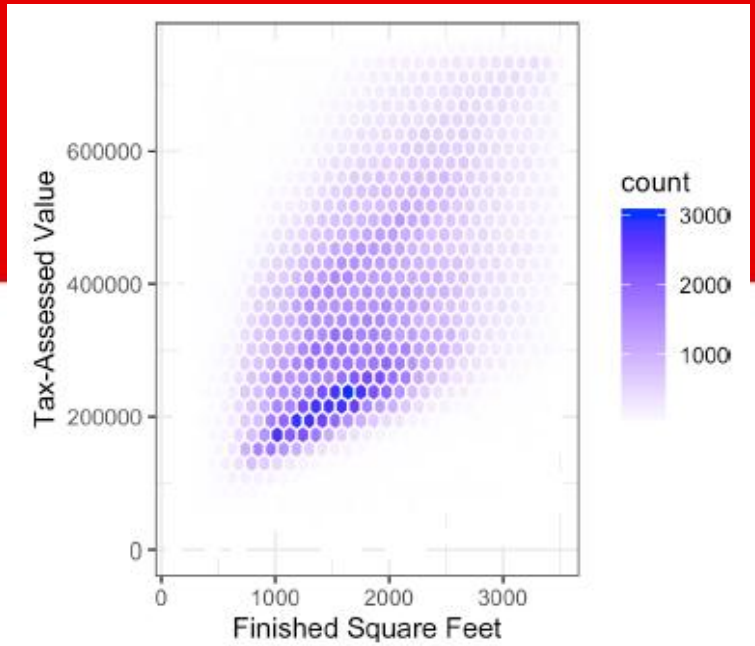
Feature Selection Goals:

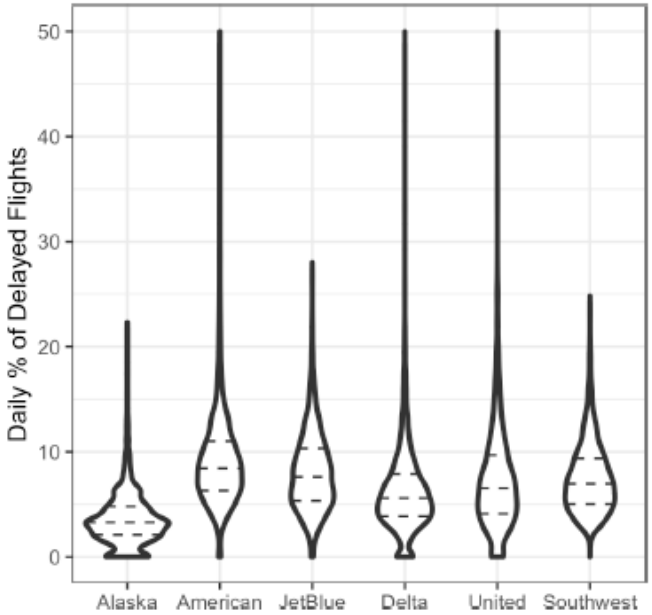
- **Low Correlation Among Features (X):**
  - Reduces multicollinearity.
  - Improves model stability and interpretability.
- **High Correlation Between Features (X) and Target (y):**
  - Enhances predictive power.
  - Ensures features are relevant to the target variable.

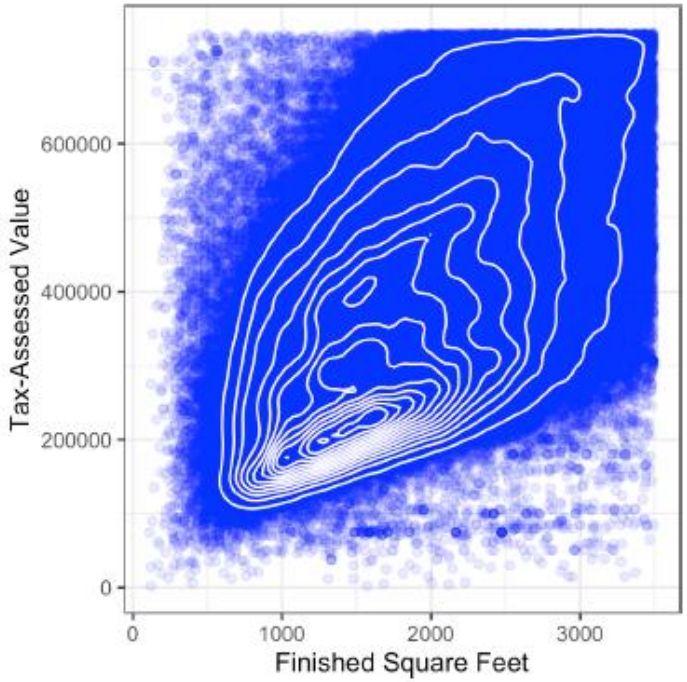
	User Rating	Reviews	Price	Year
User Rating	1.000000	-0.001729	-0.133086	0.242383
Reviews	-0.001729	1.000000	-0.109182	0.263560
Price	-0.133086	-0.109182	1.000000	-0.153979
Year	0.242383	0.263560	-0.153979	1.000000

Exploring two or more variables

- Contingency table
- Hexagonal binning
- Contour plot
- Violin plot







Outliers

What are Outliers?

- Data points significantly different from others in a dataset.
- Can indicate variability, errors, or unique phenomena.

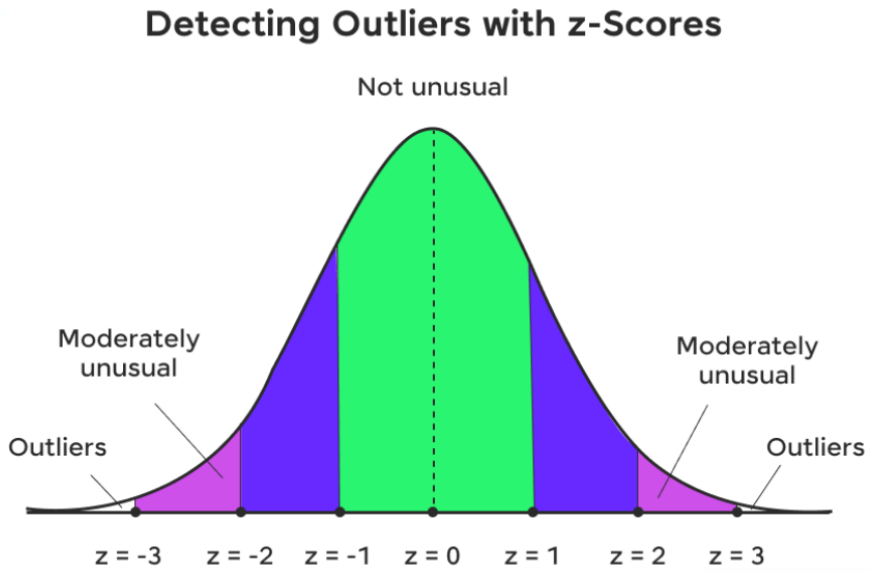
How to Detect Outliers?

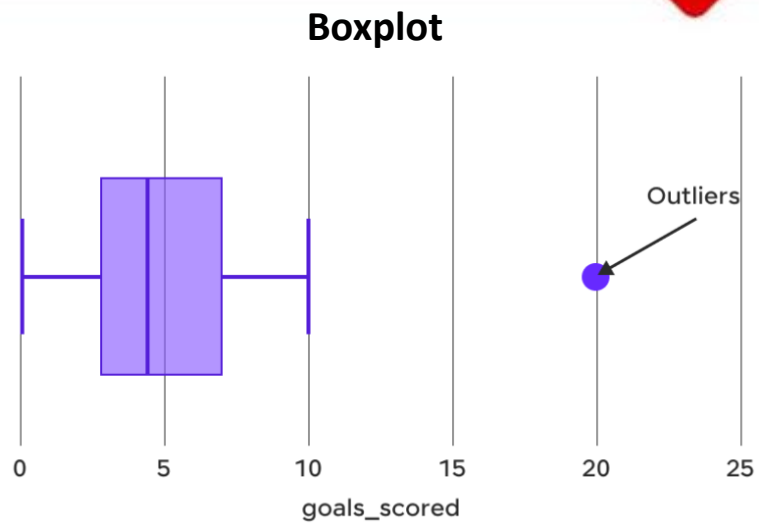
- Visual Methods: Scatter plots, box plots, …
- Statistical Methods: Z-scores, IQR (Interquartile Range), …
- Algorithmic Methods: anomaly detection

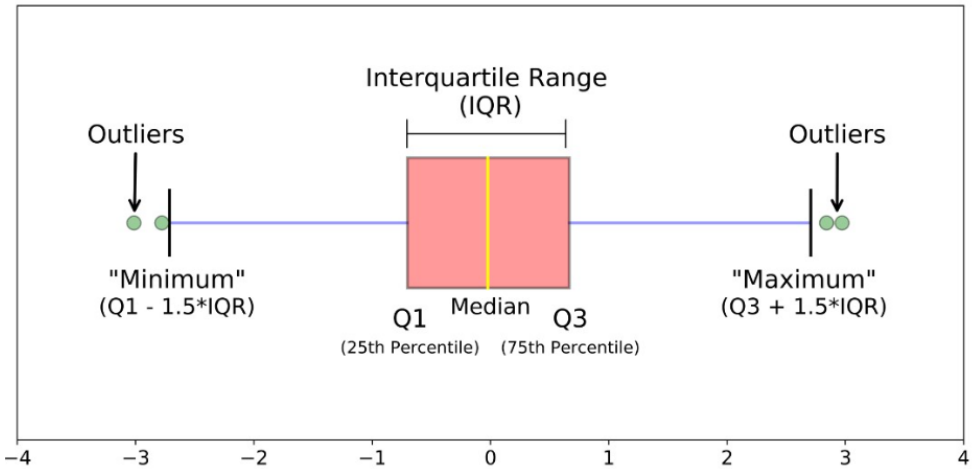
How to Deal with Outliers?

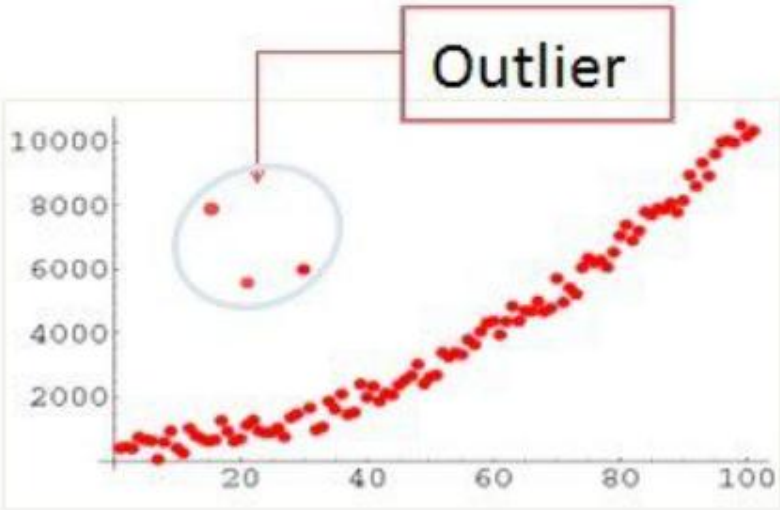
- Investigate: Determine if they are errors or valid data.
- Remove: Exclude if they are errors or irrelevant.
- Transform: Apply log transformation or normalization.
- Separate Analysis: Analyze outliers separately if they represent important phenomena.

How to detect outliers?

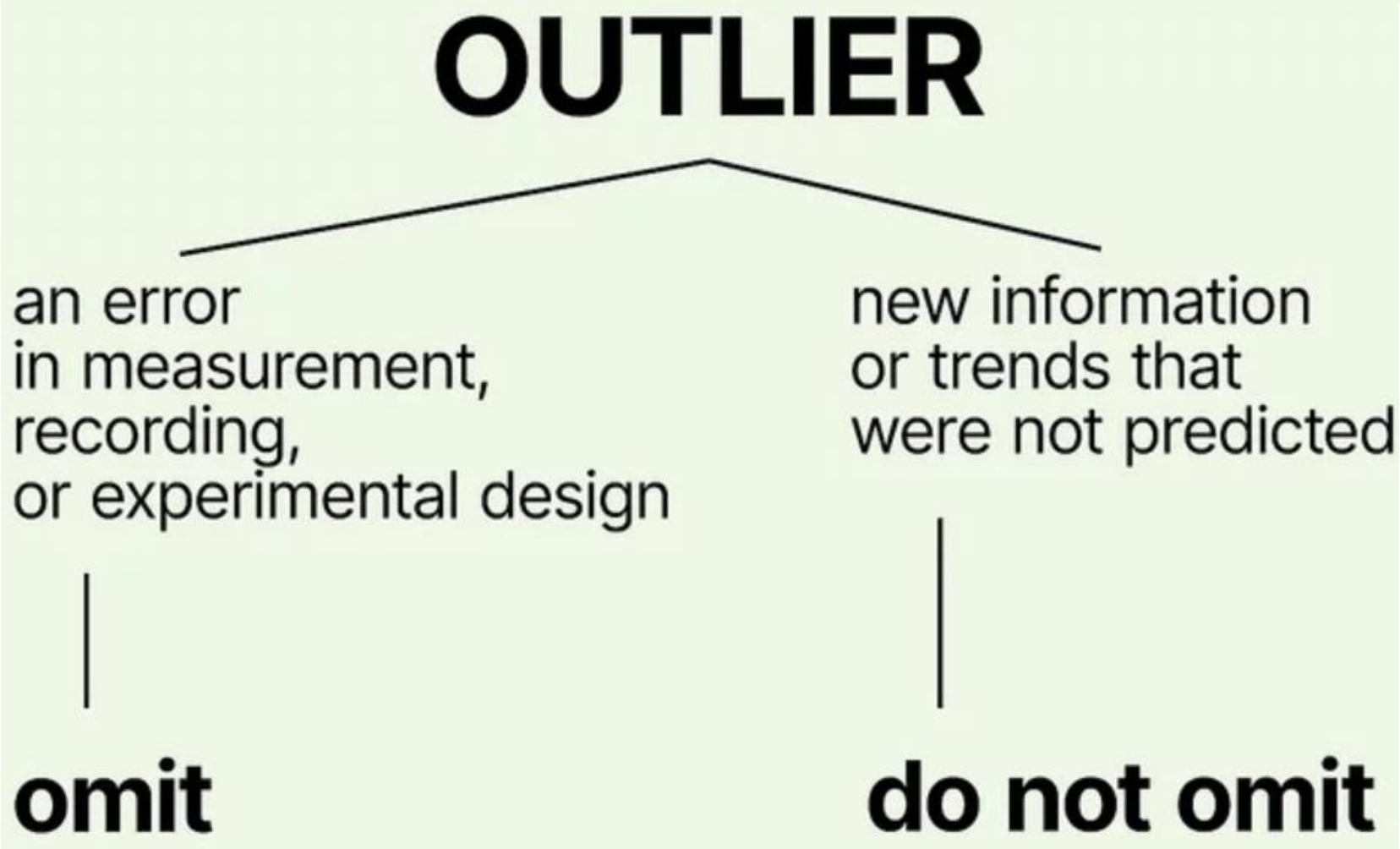








How to deal with outliers?



Handling missing values

	Id	Gender	Color
0	1	M	NaN
1	2	M	Red
2	3	F	Blue
3	4	NaN	Red
4	5	NaN	NaN
5	6	F	Red
6	7	M	Green
7	8	F	NaN

1. **Identify** missing data
2. **Dealing with** missing values:
   - **Remove** rows or columns
   - **Data imputation:**
     - Mean/Median Imputation: Replace missing values with the mean or median of the column.
     - Mode Imputation: Use the most frequent value for categorical data.
     - Forward/Backward Fill: Propagate the next or previous value.
     - Interpolation: Estimate missing values based on other data points.
     - Model-Based Imputation: Predict missing values using models like k-NN or regression.

Common pitfalls

- Ignoring missing data
- Misinterpreting data and variable types
- Assuming normality
- Overlooking patterns
- Not taking into account the units of variables

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Conclusions

- The key idea of EDA is that the **first** and most important step in any project based on data is to **look at the data**.

- By summarizing and visualizing the data, you can **gain** valuable **intuition and understanding** of the project.

- **EDA is essential.** It sets the foundation for trustworthy models and meaningful insights.

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